SAMPLE

Cosmic Explorers: The Search for Life

Are We Alone?

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Introduction

The search for life is the attempt to answer a question older than science itself: is the living Earth unique, or is life a common outcome of the universe's chemistry. Modern research pursues that question along three roads at once. Planetary scientists look for life, or its fossil traces, in our own solar system, on Mars, in the oceans hidden under the ice of Europa and Enceladus, in the organic haze of Titan. Astronomers look for the chemical fingerprints of biology in the atmospheres of planets around other stars, a young discipline transformed by the discovery of thousands of exoplanets. And the search for extraterrestrial intelligence, SETI, listens for evidence of technology, in radio signals, in laser pulses, in any sign that someone else is doing what we have only just begun to do.

This book follows all three roads. The early chapters set out the history of the idea and the science that underpins it: how the question moved from philosophy to experiment, what biology says life requires, and how scientists distinguish a genuine signature of life from the many non-living processes that can imitate one. The middle chapters describe the observation campaigns, the rovers and orbiters and telescopes that do the actual looking, and the candidate worlds where the looking is most intense. The later chapters carry the story through the research programs, the discoveries and the enduring mysteries, including the Fermi paradox, the silence that gives the search its tension, and the future missions that will define the next decades.

Two convictions organize what follows. The first is that the search is legitimate science: its hypotheses are testable, its instruments are improving by orders of magnitude, and its null results carry information. The second is that it is a search in the early stages, closer to the first decades of microbiology than to the last decades of any settled field. The reader should come away with a clear picture of what is known, what is merely plausible, and what the next generation of instruments might actually settle, a landscape changing fast enough that no book on the subject stays current for long.

One feeling belongs in place before the science begins: the question's standing in the species' mind. The search for life is the only scientific program that the whole public understands, funds, and roots for without a briefing, because the question it asks, whether anyone else is there, is one every child asks on first seeing the night sky; the modern search is not a single hunt but an organized family of them, the surveys for planets, the probes of the ocean moons, the listening posts, the telescopes reading atmospheres, and the book's subject is the family as a whole, one question with a dozen scientific addresses.

The search's two meanings also deserve their separation up front, and the book will keep them distinguished throughout. Searching for life means, first, life of any kind, a microbe in a moon's ocean or a chemical signature in a planet's air, the question the near-term missions ask; and it means, second, intelligence, company, another mind, the question the listening projects ask. The reader should hold the two apart, since the first may be answered within the reader's lifetime and the second may never be answered at all, and the book's chapters, the habitable zones, the biosignatures, the experiments, are organized by the first meaning, with the second kept honestly in its own place.

Third, the method, said once and then simply practiced: the chapters rank what is known honestly, and that ranking recurs throughout the chapters. The chapters are organized as the search is, from the assumptions to the instruments to the addresses to the open files, and the book's discipline is the distinction between what is known, what is planned, and what is merely hoped; the three tenses matter, since the search for life is the scientific subject most vulnerable to tense-blurring, the field's advocates selling the future tense and its skeptics freezing the past one, and the book's contract with the reader is to keep every claim in its right tense throughout.

Chapter 01: The Discovery of The Search for Life

The idea of other worlds is ancient. Greek atomists argued for infinite worlds, and when Copernicus in 1543 removed the home world from the cosmos's middle, the question became physical: if the Earth is a planet, the planets may be earths. Galileo saw mountains on the Moon in 1609, and Christiaan Huygens, writing in the 1690s, argued at length that the planets are inhabited. Giordano Bruno had already been burned in 1600 for, among other beliefs, proposing infinite inhabited worlds. The conviction was more theological than empirical, but it established the intuition that would drive four centuries of speculation: a large universe waste on a single living speck seemed unlikely.

The modern observational phase began badly and instructively. In 1877, Giovanni Schiaparelli mapped thin lines on Mars, canali, channels, and Percival Lowell transformed the translation error into a civilization of dying Martians

and irrigated canals, an idea the public embraced and astronomers could not confirm. The Mariner 4 flyby in 1965 returned a cratered, airless surface, and the canals evaporated: the lesson, that wishful reading can survive ordinary evidence, still disciplines the field.

The scientific search for life began almost simultaneously with the planetary disappointment. In the 1920s, Alexander Oparin and J. B. S. Haldane independently proposed that life arose from chemistry on the early Earth, making biology an outcome of physics and chemistry rather than a separate miracle. In 1952, Stanley Miller and Harold Urey ran the famous experiment, passing electric sparks through a flask of simulated early atmosphere and water, and produced amino acids, the building blocks of proteins, from non-living inputs. In 1960, Joshua Lederberg coined the term exobiology, and Frank Drake pointed a radio telescope, Project Ozma, at two nearby stars, the first modern SETI observation. The two traditions, laboratory and telescope, have defined the field ever since.

The solar-system search arrived with the spacecraft. In 1976, the two Viking landers carried biological experiments to Mars; one, the labeled release experiment, gave a result its designers debated for decades, though the consensus reading is that the reactivity was chemical, not biological, and the mission's biology package is remembered mostly as a lesson in the difficulty of unambiguous detection. In 1996, the meteorite ALH84001, a piece of Mars recovered from Antarctica, was announced to contain possible fossil traces; the claim was contested and did not survive the skeptical scrutiny, but it founded the modern science of Martian meteorites and trained a generation in biosignature debate. In 1995, the first planet around a sun-like star, 51 Pegasi b, was found, and with Kepler, launched in 2009, the exoplanet count rose into the thousands. In 2005, Cassini's instruments tasted the plume of Enceladus and found salt water and organics; in 2017, seven Earth-sized planets were announced around the star TRAPPIST-1. The search had found its places, and each of them is the subject of the chapters that follow.

The discovery chapter ends on the field's true beginning, and the dating deserves to be correct. The search did not begin with the telescope; it began with the acceptance, in the last century, that life is chemistry, that the living cell is a running reaction rather than a vital essence, since chemistry can happen anywhere the conditions allow; the origin-of-life experiments, the famous flask that made the amino acids from simple gases, deserve credit as the search's founding demonstration, the proof that the ingredients are ordinary, and the modern program, planet-hunting and moon-probing alike, is the search for where that ordinary recipe has been allowed to run.

The found worlds' demographics close the discovery story, the fact that turned the search from philosophy into fieldwork. The first planet around another sun was confirmed in the 1990s, and the catalogs since then have grown to thousands, with the statistics implying that most stars host planets, and the Milky Way accordingly holds more worlds than the species has ever had names for; the demographic shift reads as the search's hinge, since before the planets were found the question was whether there were any addresses, and now the question is which of the billions of addresses to check first.

The radio era produced its own prehistory, mostly comic and occasionally sober. The inventors of wireless telegraphy speculated in the press that their new waves might reach Mars as easily as they reached ships at sea, and the newspapers of the 1890s through the 1920s periodically reported signals from the neighbor world, none of which survived inspection. The high point of the era's seriousness came at a favorable opposition of Mars in the early 1920s, when a military radio chief reportedly ordered a nationwide quiet period so that a leading astronomer could listen for Martian broadcasts with the still-primitive receivers of the day; nothing was heard, the episode is preserved mainly as an anecdote about pre-disciplinary innocence, and the modern field cites it as the era when the question was asked by instruments for the first time, with the wrong instruments and no method.

The exoplanet hinge deserves its own two dates. The first planet found orbiting a sun-like star, a hot Jupiter racing around its home in a few days, arrived in 1995 and won its discoverers the Nobel Prize a quarter-century later, and the method that found it, the wobble of a star tugged by its planet's mass, immediately gave the field a second leg beside the transits. The space telescope that followed, watching one crowded star field continuously for four years, pushed the confirmed count into the thousands and established the occurrence rates the search now plans around; the census's standing irony is that the method finds what blocks light, and so the catalog is dominated by worlds built to be found, the close, the large, and the lucky-aligned, with the quiet Earth-like cases hidden in the statistics until the next generation of instruments.

The discipline's institutionalization arrived with a controversy. The Martian meteorite affair of the 1990s, a claim about microfossils in a rock from the Antarctic ice, forced the field's elders to build what the controversy exposed as missing: a peer-review home for the work, a funding line for the laboratories, and a national center to coordinate

them, and the astrobiology institutes founded in the aftermath institutionalized the field under its current name, wider than the old exobiology, reaching from prebiotic chemistry to the deep biosphere to the exoplanet censuses. The naming matters to the discovery story because it marks the moment the search stopped being an interest of a few laboratories and became a discipline with students, curricula, and standing committees, and the generation trained inside that discipline is the one now flying the missions.